



Project (1)

Learning objectives: by the end of this project, you should be able to:

- **LO1:** Describe the electrical behavior of the MOS transistor in scaled technologies taking into consideration short channel effects
- **LO2:** Design the region of inversion (weak, moderate, strong) and level of saturation (triode, saturation) of each transistor in your circuit to meet the requirements of your application
- **LO3:** Size optimally all the transistors in any advanced technology node using the gm/ID methodology

You can work in groups of max. 2 or alone, as you prefer.

Resources: you can discuss the progress of your project with us during the flipped classrooms on Tuesday 15/3 and Thursday 17/3.

Deadline: Tuesday 22/3 before class



Project (2)

Project description: Your boss saw the design example on this gm/ID lecture. She says that this design is pretty close to what your company needs for a product but wants to enlist your skills for an improved version that meets the following *objectives*:

- The device width must be minimized as much as possible given a drain current budget of 500 μA .
- The settling time must be below 5.5 ns with self-loading included.
- Substitute the ideal current source with a PMOS and optimize the ratio g_{mp}/g_{mn} to maximize your dynamic range $DR = \frac{P_{out}}{P_{n,out}}$ (assume sinusoidal signal for P_{out}).
- All other parameters (FO, G, etc.) remain the same.
- Also, all key specs (settling time, static error, total integrated noise) should be validated in HSpice and must meet the target values.
- One way to solve this problem is to randomly tweak the circuit parameters across a huge number of HSpice runs, but we would like you to use this problem as an opportunity to embrace a more systemic design approach with analytical backing. That said, there is no unique solution to this problem. We suggest that you use the starter file "igs_cap_sizing.m" and modify it as you see fit. For example, you could add for loops to iterate, or change the design flow entirely (feel free to be creative!).



Project (3)

Deliverables: you will need to submit the following material on Brightspace by the deadline (22/03/2022 at 13:30)

- A PDF report of your final design following the IEEE Transactions format on Brightspace with max. 4 pages (**NB: reports that do not follow this format or exceed the max. number of pages will not be graded**). You should include the following in your report:
 - A schematic of your final design.
 - A short description of your optimization approach with your final design parameters (use the Table template in these slides for this). The key objective is to explain how you reached your design point.
 - A discussion on the comparison of the key performance specs (settling time, static error, total integrated noise) computed by your code versus your HSpice results. If properly done, the discrepancies should be small (~10% or less).
 - Plots for the settling transient (see slide 35 – gm/ID part 2), settling time (see slide 36 – gm/ID part 2) and the running output noise integral (versus frequency) with annotated end value in μV_{rms} (the starter file has some code for this)
- The MATLAB code used for your project. This code should be self-contained and error free. We will run the code ourselves during grading.
- The LTSpice files used for your project. Again, we will check the simulation results during grading, so make sure that all files run properly before submission.



Project (4)

Item	Value
W [μm]	
L [μm]	
Area [μm^2]	
CR	
gm/ID [S/A]	
Input Noise [μV_{rms}]	
Settling Time [ns]	
Static Error [%]	
ID [μA]	



Project (5)

Rubric: the grade will be based on the following aspects of the project

- **Design Specifications [settling time, static error, noise] (20%)**
 - All requirements are met, and DR is maximized
- **MATLAB Script – gm/ID loop (20%)**
 - Loop is implemented without errors
- **MATLAB Script – sizing loop (20%)**
 - Loop is implemented without errors
- **LTSpice Simulations (10%)**
 - Fine-tuning is implemented to obtain best-in-class specs, and all simulations are correctly set
- **Design Methodology (30%)**
 - Design choices are motivated (e.g., gm/ID level), showing a clear understanding of MOS physics. Discrepancies between MATLAB and LTSpice are discussed thoroughly.